

Lesson 07: Linear Second-Order ODEs - Homogeneous Case

Professor Dave Explains Differential Equations Master Series · Lesson 7

Video: Linear Second-Order Differential Equations Part 1: Homogeneous Case Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that second-order linear differential equations are the heartbeat of engineering—they govern mass-spring-damper suspensions, RLC radio tuning circuits, and seismic building vibrations. For constant coefficients $ay'' + by' + cy = 0$, we test the exponential ansatz $y = e^{rx}$, converting calculus into simple quadratic algebra $ar^2 + br + c = 0$. The roots of this characteristic equation govern whether the physical system decays monotonically, critically returns to rest, or oscillates sinusoidally.

2. Core Theoretical Framework & Rigorous Formulations

The 3 Characteristic Root Cases

Case	Roots of $ar^2 + br + c = 0$	General Complementary Solution $y_c(x)$	Physical Behavior
Case 1	Distinct Real $r_1 \neq r_2$	$y_c = c_1 e^{r_1 x} + c_2 e^{r_2 x}$	Overdamped non-oscillatory decay
Case 2	Repeated Real $r_1 = r_2 = r$	$y_c = c_1 e^{rx} + c_2 x e^{rx}$	Critically damped rapid return
Case 3	Complex Conjugates $\alpha \pm i\beta$	$y_c = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$	Underdamped sinusoidal oscillation

The Wronskian Determinant: $W(y_1, y_2) = y_1 y_2' - y_1' y_2 \neq 0$ proves linear independence.

3. Fully Solved Whiteboard Problems

Classroom Problem: Complete Solution of an Underdamped Oscillator IVP

Problem Statement: Solve the initial value problem $y'' + 4y' + 13y = 0$, with $y(0) = 2$ and $y'(0) = 1$.

- Step 1: Form Characteristic Auxiliary Equation
- $r^2 + 4r + 13 = 0$.
- Step 2: Solve Quadratic Roots
- $r = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = -2 \pm 3i$. Here $\alpha = -2$ and $\beta = 3$.
- Step 3: Write General Solution via Euler's Formula
- $y(t) = e^{-2t} (c_1 \cos(3t) + c_2 \sin(3t))$.
- Step 4: Enforce Initial Displacement $y(0) = 2$
- $y(0) = e^0 (c_1 \cos 0 + c_2 \sin 0) = c_1 \implies c_1 = 2$.
- Step 5: Differentiate and Enforce Initial Velocity $y'(0) = 1$
- $y'(t) = -2e^{-2t} (2 \cos 3t + c_2 \sin 3t) + e^{-2t} (-6 \sin 3t + 3c_2 \cos 3t)$. At $t = 0$: $y'(0) = -2(2) + 3c_2 = -4 + 3c_2 = 1 \implies 3c_2 = 5 \implies c_2 = \frac{5}{3}$.
- Step 6: Assemble Final Response
- $y(t) = e^{-2t} (2 \cos(3t) + \frac{5}{3} \sin(3t))$.

⚠ Common Exam Pitfall & Trapdoor Avoidance

When differentiating $y(t) = e^{\alpha t} (c_1 \cos \beta t + c_2 \sin \beta t)$ to enforce velocity conditions, ALWAYS apply the Product Rule carefully. Missing the derivative of the exponential envelope is a chronic error.

4. Key Takeaways & Exam Strategy Formula Card

🔑 Core Takeaway & Exam Rule

Complex roots $\alpha \pm i\beta$ yield decaying sinusoidal oscillations with envelope $e^{-\alpha t}$ and frequency β .

- Video reference: [Linear Second-Order Differential Equations Part 1: Homogeneous Case](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.